

Ziyang Song

CONTACT INFORMATION

Stocker Center 313
School of Electrical Engineering and Computer Science
Ohio University
Athens, Ohio, United States

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Website
LinkedIn
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RESEARCH INTERESTS

My research focuses on AI for health, with the goal of developing trustworthy and translational AI to address biomedical challenges.

1. **Trustworthy LLMs in Healthcare.** I develop methods to improve explainability, uncertainty quantification, and risk control for biomedical LLMs and multi-agent clinical systems.
2. **Foundation Models for Longitudinal Health Data.** I design foundation models for irregular and longitudinal healthcare data, including electronic medical records, laboratory measurements, and biosignals.
3. **Interpretable AI for Population Health.** I develop probabilistic models and scalable statistical inference methods to learn interpretable representations from electronic health records and genotype data, supporting phenotyping, incidence prediction, and gene discovery.
4. **Generative AI for Biology.** I develop foundation models for irregular single-cell RNA-seq time-series data and study domain adaptation methods that leverage unlabeled data.

FULL-TIME PROFESSIONAL EXPERIENCE

Ohio University
School of Electrical Engineering and Computer Science
Assistant Professor

Athens, OH, United States

Aug. 2025 – Present

EDUCATION

McGill University
Ph.D. in Computer Science

- **Affiliation:** Mila – Quebec AI Institute
- **Advisor:** Yue Li

Montreal, QC, Canada

Sep. 2020 – May 2025

Concordia University
M.Sc. in Quality Systems Engineering

- **Advisor:** Nizar Bouguila

Montreal, QC, Canada

Sep. 2018 – Nov. 2019

University of Shanghai for Science and Technology
B.Eng. in Computer Science

Shanghai, China

Sep. 2014 – Jun. 2018

PUBLICATIONS

Note: [†] equal contribution; * corresponding authors.

Conference Papers:

1. Xinyu Wang, Hanwei Wu, Zhenghan Tai, Sicheng Lyu, Qincheng Lu, Ziyu Zhao, Jijun Chi, Jingrui Tian, Xiao-Wen Chang, **Ziyang Song**^{*}. SafeRx-Agent: A Knowledge-Grounded Multi-Agent Framework for Safe and Explainable Medication Recommendation. *EMNLP Main Conference*, 2026.
2. **Ziyang Song**, Qincheng Lu, Hao Xu, He Zhu, David Buckeridge, Yue Li. TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *ACM-BCB*. 2024. **Rising Star Award**.
3. Zilong Wang, Ruohan Wang, **Ziyang Song**, David Buckeridge, Yue Li. MixEHR-Nest: Identifying Subphenotypes within Electronic Health Records through Hierarchical Guided-Topic Modeling *ACM-BCB*. 2024.

4. **Ziyang Song**, Qincheng Lu, He Zhu, David L. Buckeridge, Yue Li. Bidirectional Generative Pre-training for Improving Healthcare Time-series Representation Learning. *Machine Learning for Healthcare Conference (MLHC)*, PMLR. 2024.
5. **Ziyang Song**, Yuanyi Hu, Aman Verma, David L. Buckeridge, Yue Li. Automatic Phenotyping by a Seed-guided Topic Model. *KDD*. 2022. **Health Day Best Paper Award**.
6. **Ziyang Song**, Xavier Sumba Toral, Yixin Xu, Aihua Liu, Liming Guo, Guido Powell, Aman Verma, David Buckeridge, Ariane Marelli, Yue Li. Supervised Multi-Specialist Topic Model With Applications on Large-Scale Electronic Health Record Data *ACM-BCB*. 2021.
7. **Ziyang Song**, Ornela Bregu, Samr Ali, Nizar Bouguila. Variational inference of finite asymmetric gaussian mixture models. *IEEE SSCI*. 2019.
8. **Ziyang Song**, Samr Ali, Nizar Bouguila. Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction. *ICIAR*. 2019.

Journal Papers:

1. Ziqi Yang[†], **Ziyang Song**[†], Shadi Zabad, Marc-André Legault, Yue Li. PheCode-guided multi-modal topic modeling of electronic health records improves disease incidence prediction and GWAS discovery from UK Biobank. *Briefings in Bioinformatics*. 2026.
2. **Ziyang Song**, Qincheng Lu, He Zhu, David Buckeridge, Yue Li. TrajGPT: Irregular Time-Series Representation Learning of Health Trajectory. *IEEE Journal of Biomedical and Health Informatics (J-BHI)*. 2025.
3. **Ziyang Song**, Qincheng Lu, Hao Xu, Ziqi Yang, He Zhu, David Buckeridge, Yue Li. Time-lyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare. *Health Information Science and Systems*. 2025.
4. Yuesong Zou, Ahmad Pesaranghader, **Ziyang Song**, Aman Verma, David Buckeridge, Yue Li. Modeling electronic health record data using an end-to-end knowledge-graph-informed topic model. *Scientific Reports*. 2022.
5. **Ziyang Song**, Samr Ali, Nizar Bouguila. Bayesian inference for infinite asymmetric gaussian mixture with feature selection. *Soft Computing*. 2021.
6. **Ziyang Song**, Samr Ali, Nizar Bouguila. Background subtraction using infinite asymmetric Gaussian mixture models with simultaneous feature selection. *IET Image Processing*. 2020.
7. **Ziyang Song**, Samr Ali, Nizar Bouguila, Wentao Fan. Nonparametric hierarchical mixture models based on asymmetric gaussian distribution. *Digital Signal Processing*. 2020.

Preprints & Under Submission:

1. Weihao Li, Jun Bai, **Ziyang Song***. FedAgentKE: Federated Semantic Knowledge Evolution for Heterogeneous Agents. <https://arxiv.org/abs/2607.21361>.
2. Bo-Hong Wang, Baicheng Peng, Ruilin Wang, Jun Bai, **Ziyang Song**, Yue Li. ChatHealthAI: Aligning Electronic Health Record Representations with Large Language Models for Grounded Clinical Reasoning. <https://arxiv.org/abs/2606.02802>.
3. Ruilin Wang, Bo-Hong Wang, Elizabeth Kourbatski, Jun Bai, Hegang Chen, **Ziyang Song**, Gilles Boire, Marie Hudson, Yue Li. DoctorAgents: an agentic framework to iteratively refine AutoML pipeline for small clinical temporal data. <https://arxiv.org/abs/2608.05375>. Submitted to *KDD*, under review.
4. Yixuan Li, Weihao Li, **Ziyang Song***. HCOE: Hyperbolic Clinical Ontology Embeddings from Biomedical Language Models. Submitted to *IEEE BIBM*, under review.
5. Xinyu Wang, Yixuan Li, Hanwei Wu, Qincheng Lu, Chi-Kuang Yeh, Xiao-Wen Chang, **Ziyang Song***. Patients-like-me: A Variational LM-GNN Framework for Explainable Clinical Prediction. Submitted to *EACL*, under review.

6. Yu Gu, Zijun Yu, Chi-Kuang Yeh, Xinyu Wang, **Ziyang Song***. Severity-Controlled Prediction Sets for Medication Recommendation. Submitted to *NeurIPS*, under review.
7. Yiran Ding, Yixuan Li, **Ziyang Song***. scUFC: Single-Cell Unsupervised Fine-Tuning via Clustering for Foundation Models. Submitted to *NeurIPS*, under review.
8. **Ziyang Song***, Lia Shen, Yixuan Li, Qincheng Lu, Weihao Li, Haoran Xu, He Zhu, Xinyu Wang, Yue Li *. SMI: Semantic Medical ID for Hierarchy-Aware Concept Representation. Submitted to *NeurIPS*, under review.
9. Jun Bai, **Ziyang Song**, Yue Li. FedEHR-GEN: Federated Synthetic Time-Series EHR Generation via Latent Space Alignment and Distribution-Aware Aggregation. Submitted to *IEEE J-BHI*, under review.

INVITED TALKS &
PRESENTATIONS

Invited Talks

OSC Research Symposium & NSF AI workshop	<i>Toward Trustworthy LLMs in Healthcare with Medical Ontologies.</i>	Apr. 2026
Global AI Youth Forum	<i>Toward Trustworthy LLMs in Healthcare with Medical Ontologies.</i>	Mar. 2026
Northern Arizona University	<i>Representation Learning of Electronic Health Records.</i>	Mar. 2025
San José State University	<i>Representation Learning of Electronic Health Records.</i>	Mar. 2025
Ohio University	<i>Representation Learning of Electronic Health Records.</i>	Feb. 2025
South Dakota State University	<i>Representation Learning of Electronic Health Records.</i>	Feb. 2025
University of Texas at El Paso	<i>Representation Learning of Electronic Health Records.</i>	Feb. 2025
Microsoft Research Asia	<i>Time-series Transformer for Representation Learning and Health Forecasting</i>	Nov. 2024

Guest Lectures

Ohio University	<i>Generative AI for Healthcare.</i>	Nov. 2025
McGill University	<i>Time-series Transformer in Electronic Health Records.</i>	Nov. 2024
McGill University	<i>Time-series Transformer in Electronic Health Records.</i>	Nov. 2023

Oral Presentations

Canadian Society for Epidemiology and Biostatistics (CSEB)	<i>AI-driven approach for computational disease phenotyping within the CARTaGENE cohort.</i>	Aug. 2025
ACM BCB	<i>TimelyGPT: Extrapolatable Transformer Pre-training for Long-term Time-Series Forecasting in Healthcare.</i>	Nov. 2024
Scientific Meeting of the Canadian Translational Geroscience Network	<i>AI-driven approach for computational phenotyping with CARTaGENE cohort.</i>	Sep. 2024
Tokyo Symposium on Genomic Medicine, Therapeutics and Health	<i>Tutorial: Phenotyping and PheWAS using MixEHR-Seed.</i>	Apr. 2024
ISMB GLBIO	<i>MixEHR-SAGE: A seed-guided topic model to improve phenotyping for PheWAS analysis in UK Biobank data.</i>	May. 2023
KDD	<i>Automatic phenotyping by a seed-guided topic model.</i>	Aug. 2022
ACM BCB	<i>Supervised multi-specialist topic model with applications on large-scale electronic health record data.</i>	Aug. 2021

Poster Presentations

Global AI Youth Forum	<i>Toward Trustworthy LLMs in Healthcare with Medical Ontologies.</i>	Mar. 2026
NeurIPS 2025 Responsible Foundation Model Workshop	<i>SMI: Semantic Medical ID for Hierarchy-Aware Concept Representation.</i>	Nov. 2025
NeurIPS 2024 TSALM Workshop	<i>TrajGPT: Healthcare Time-Series Representation Learning for Trajectory Prediction.</i>	Dec. 2024
MLHC	<i>Bidirectional generative pre-training for improving healthcare time-series representation learning.</i>	Aug. 2024
50th Anniversary of SOCS at McGill University	<i>MixEHR-Seed: Automatic phenotyping by a seed-guided topic model.</i>	Oct. 2022
ICIAR	<i>Bayesian learning of infinite asymmetric gaussian mixture models for background subtraction.</i>	Aug. 2019

AWARDS, GRANTS, AND FUNDING

As Principal Investigator (Aug. 2025 onward)

Human-first AI Initiative Faculty Fellow, Ohio University	<i>Fellowship</i>	2026–2027
Tinker Research Grant, Thinking Machines Lab	<i>\$5,000 Credits</i>	Aug 2026
Google Cloud Research Credits	<i>\$5,000 Credits</i>	May 2026
NVIDIA Academic Grant Program Award	<i>Gift</i>	Dec. 2025

Prior to Principal Investigator Role (Before Aug. 2025)

FRQNT Provincial Doctoral Research Scholarship	<i>Scholarship</i>	May. 2023–Sep. 2025
ACM BCB Rising Star Award	<i>Award</i>	Nov. 2024
KDD Health Day Best Paper Award	<i>Award</i>	Aug. 2022

TEACHING EXPERIENCE	Ohio University	
	Instructor	Fall 2026
	<i>AI 5010/4010 Foundations of Deep Learning</i>	
	Instructor	Spring 2026
	<i>AI 3300 Statistical Learning</i>	
	Instructor	Fall 2025
	McGill University	
	Teaching Assistant	Winter 2023
	<i>COMP 551 Applied Machine Learning</i>	
	Teaching Assistant	Fall 2022
	<i>COMP 565 Machine Learning in Genomics and Healthcare</i>	
INTERNSHIP EXPERIENCE	Centre Hospitalier Universitaire Sainte-Justine (CHU-SJ)	Montreal, QC, Canada
	<i>Research Intern</i>	Jun. 2023–Dec. 2024
	Developed a clinical decision-support system using a probabilistic model to identify high-risk patients who may be missed by existing rule-based screening methods.	
	Nanyang Technological University	Singapore
	<i>International Research Intern</i>	Jan. 2020–Jun. 2020
	Developed probabilistic models with stochastic optimization to analyze retail data, improving personalized product recommendations and targeted promotions.	
SUPERVISION & MENTORING	Ohio University (Assistant Professor)	
	Ph.D. Students	
	Xuanlu Zhang	Sep. 2026–Present
	Shunning Liang Tan	Sep. 2026–Present
	Undergraduate Students	
	Sheil Iyer	Sep. 2025–Present
	<i>Topic/Output:</i> Agent with memory to unify email, Calendar, Notion, and Slack.	
	Research Assistant Students	
	Hanwei Wu (M.Sc., McMaster University)	May. 2025–Present
	<i>Topic/Output:</i> Graph explainer for biomedical LLMs; Submitted to ICML 2026.	
	Zihao Li (M.Sc., Northwestern University)	Jul. 2025–Present
	<i>Topic/Output:</i> Semantic IDs in biomedical LLMs; NeurIPS Workshop 2025.	
	Yiran Ding (M.Sc., Brown University)	Jul. 2025–Present
	<i>Topic/Output:</i> Representation clustering for biomedical LLMs.	
	Xiaohang Xie (B.S., OSU)	Jul. 2025–Present
	<i>Topic/Output:</i> Confidence calibration for biomedical LLMs.	
	Yang Hu (B.S., OSU)	Jul. 2025–Present
	<i>Topic/Output:</i> Hyperbolic space for biomedical LLMs.	
	Jiahe Shen (M.Sc., UW–Madison)	Jul. 2025–Dec. 2025
	<i>Topic/Output:</i> Semantic IDs in biomedical LLMs; NeurIPS Workshop 2025.	
	<i>Placement:</i> Research intern, MD Anderson Cancer Center.	
	McGill University (Ph.D. Mentor)	

Ruilin Wang (M.Sc., McGill University)	Sep. 2024–May. 2025
<i>Topic/Output:</i> LLM agent. Submitted to ICML 2026.	
Ziqi Yang (B.S./M.Sc., McGill University)	Sep. 2022–May. 2025
<i>Topic/Output:</i> Expert-guided topic model for GWAS; Briefings in Bioinformatics.	
<i>Placement:</i> M.Sc. student, McGill University.	
Ruohan Wang (B.S./M.Sc., McGill University)	Sep. 2022–Sep. 2024
<i>Topic/Output:</i> Expert-guided topic model; ACM-BCB 2024.	
<i>Placement:</i> Ph.D. student, Brown University.	
Hao Xu (B.S., McGill University)	May. 2023–Sep. 2023
<i>Topic/Output:</i> Time-series Transformer; ACM-BCB 2024.	
<i>Placement:</i> Ph.D. student, University of Pennsylvania.	
Ziyu Zhao (B.S., McGill University)	May. 2023–Sep. 2023
<i>Topic/Output:</i> Time-series Transformer.	
<i>Placement:</i> M.Sc. student, McGill University.	
Yuanqi Hu (B.S., McGill University)	May. 2021–May. 2022
<i>Topic/Output:</i> Expert-guided topic model; KDD 2022.	
<i>Placement:</i> M.Sc. student, Columbia University.	
Yixin Xu (B.S., McGill University)	Sep. 2020–Sep. 2021
<i>Topic/Output:</i> Multi-specialist topic model; ACM-BCB 2021.	
<i>Placement:</i> M.Sc. student, Duke University.	

SERVICE

Conference Reviewer

WSDM	2027
AAAI	2027
KDD	2024–2026
ACM CAIS	2026
NeurIPS	2024, 2026
ICML	2025–2026
ICLR	2025–2026
IJCAI	2025–2026
CROSS-LLM	2025
ACM SIGSPATIAL	2025
ML4H	2024

Journal Reviewer

Expert Systems with Applications	2026
Franklin Open	2026
IEEE-JBHI	2026
Applied Intelligence	2025 - 2026
IEEE TNNLS	2026
BMJ Digital Health & AI (expert peer reviewer)	2025–2026
TMLR	2025–2026
Big Data Mining and Analytics	2024–2025

Ohio University

Human-first AI Initiative Faculty Fellow, Ohio University	2026–2027
CS/AI Assessment & Accreditation Committee member	2025–2027
Russ College of Engineering 3 Minute Thesis (3MT) Competition faculty judge	Feb. 2026

McGill University and Mila

Evaluation committee for Mila Supervision Request Process	2024
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REFEREES

Dr. Yue Li, Associate Professor, School of Computer Science, McGill University
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Dr. David Buckeridge, Professor, School of Population and Global Health,
McGill University
Email: send.Buckeridge.BC61C28AEC2@interfoliodossier.com

Dr. Chi-Kuang Yeh, Assistant Professor, Department of Mathematics and
Statistics, Georgia State University
Email: cyeh@gsu.edu

Dr. Archer Yi Yang, Associate Professor, Department of Mathematics and
Statistics, McGill University
Email: send.Yang.BCEBBEFEC2@interfoliodossier.com